Parametric Computational Design of Kaplan Turbine Blade Profiles Using NACA Airfoils

Prof. Sachin Komble, Aarya Velankar, Tanishq Chavan, Rugwed Ushir, Krishna Vyas, Sarvesh Suryavanshi
Department of Mechanical Engineering
Vishwakarma Institute Of Technology, Pune

***Abstract*—Kaplan turbines are widely used in low-head and high-flow hydropower applications due to their ability to maintain high efficiency under varying operating conditions. The performance of these turbines is strongly influenced by the aerodynamic profile of their blades, which governs the conversion of hydraulic energy into mechanical energy. This paper presents a computational framework for the design and generation of Kaplan turbine blade profiles using NACA 4-digit airfoils implemented through Python programming.**

The proposed methodology integrates airfoil coordinate generation, hydraulic parameter calculations, and velocity triangle analysis to determine blade geometry across multiple radial sections. The blade is discretized from hub to tip, and each section is designed based on local flow conditions, resulting in a realistic twisted blade profile. The generated airfoil sections are visualized and exported into CAD-compatible formats, enabling seamless three-dimensional modeling.

The approach offers a significant reduction in design effort while maintaining accuracy and flexibility. By allowing parametric modification of input variables such as power, head, and efficiency, the model provides a practical tool for preliminary turbine design and educational purposes. The results demonstrate that the developed framework successfully captures essential characteristics of Kaplan turbine blades and can serve as a foundation for further optimization and simulation studies.

***Index Terms*—Kaplan Turbine, NACA Airfoil, Hydropower, Blade Design, Computational Modeling, Python Programming, Velocity Triangle, CAD Integration, Parametric Design**

I. Introduction

Hydropower continues to be one of the most reliable sources of renewable energy due to its ability to provide continuous and predictable power generation. Among various types of turbines, Kaplan turbines are specifically designed for low-head and high-flow conditions, making them suitable for river-based installations. Their adjustable blade mechanism allows them to maintain high efficiency even under varying operating conditions.

The performance of a Kaplan turbine is strongly influenced by the geometry of its blades. The blade functions similarly to an airfoil, generating lift forces that convert hydraulic energy into mechanical energy. Therefore, even small changes in blade shape or orientation can significantly impact efficiency, torque generation, and cavitation characteristics.

Traditional methods of blade design involve manual calculations, empirical relations, and iterative physical modeling. These approaches are time-consuming and often lack flexibility. The present work aims to address these limitations by introducing a computational method using Python, which automates the design process and allows rapid modification of parameters.

II. Theoretical Background

A. Airfoil Theory

Airfoil theory forms the basis of turbine blade design, as the blade interacts with flowing water in a manner similar to an aerodynamic wing. NACA 4-digit airfoils are widely used due to their standardized mathematical formulation and reliable performance characteristics.

The thickness distribution of a NACA airfoil is given by:

$$y_t = 5t(0.2969\sqrt{x} - 0.1260x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4) \tag{1}$$

This equation ensures a smooth and realistic profile. The camber line determines the curvature of the airfoil and directly affects lift generation. By combining thickness and camber equations, the upper and lower surfaces of the airfoil can be accurately defined.

The use of NACA airfoils in turbine design provides a balance between performance and simplicity. Their predictable behavior makes them suitable for computational modeling and engineering applications.

B. Hydraulic Design Principles

The power output of a turbine depends on flow rate and available head, and is expressed as:

$$P = \rho g Q H \eta \tag{2}$$

From this, the flow rate can be determined:

$$Q = \frac{P}{\rho g H \eta} \tag{3}$$

These parameters form the foundation of turbine design. Accurate estimation of flow rate is essential for determining blade dimensions and velocity components.

In addition to flow rate, parameters such as peripheral velocity, flow velocity, and whirl velocity are calculated. These values are used to construct velocity triangles and determine blade orientation.

C. Velocity Triangle Analysis

The velocity triangle provides a graphical representation of fluid flow relative to the turbine blade. It consists of three main velocity components: absolute velocity, relative velocity, and blade speed.

At any section of the blade, water approaches with an absolute velocity that can be decomposed into axial (flow velocity) and tangential (whirl velocity) components. The blade moves with a peripheral velocity, which depends on its radial position.

The blade angles are calculated using:

$$\tan\beta_1 = \frac{V_f}{u - V_w} \tag{4}$$

$$\tan\beta_2 = \frac{V_f}{u} \tag{5}$$

These angles ensure that the blade is aligned with the flow, minimizing energy losses. Proper design of velocity triangles is essential for achieving high efficiency and smooth operation.

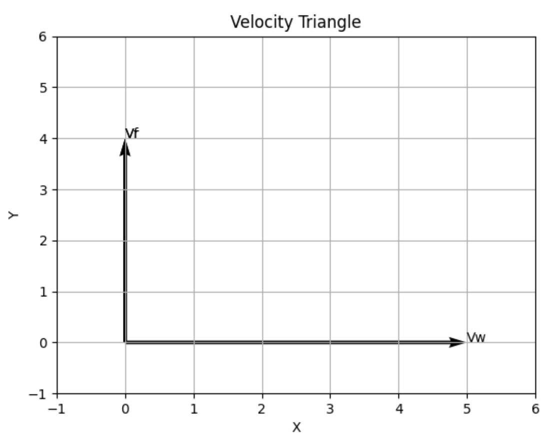

Fig. 1. Velocity Triangle at Blade Section

III. Methodology

The computational methodology begins with user-defined inputs such as power output, head, efficiency, number of blades, and NACA airfoil selection. These inputs are used to calculate flow rate and velocity parameters.

The blade is divided into multiple radial sections from hub to tip. Each section experiences different flow conditions, requiring variation in geometry. For each section, parameters such as diameter, velocity, blade angle, and chord length are calculated.

The airfoil is then scaled according to chord length and rotated based on blade angle. This transformation results in a twisted blade geometry, which is characteristic of Kaplan turbines. The generated coordinates are exported as DXF files for CAD modeling.

IV. Results

A. Input Parameters

TABLE I
Design Input Parameters

Parameter	Value
Power	100 kW
Head	50 m
Efficiency	0.9
Blades	6
Airfoil	NACA 2412

These values represent a typical operating condition for a Kaplan turbine. They were used as input for generating blade geometry.

B. Blade Section Parameters

TABLE II
Section-wise Blade Parameters

Section	Diameter	Velocity	β_1	β_2	Chord
1	0.30	8.2	42.5	28.3	0.12
2	0.40	10.1	38.2	25.6	0.10
3	0.50	12.4	34.7	22.8	0.08
4	0.60	14.0	31.5	20.1	0.06
5	0.70	16.2	28.9	18.0	0.05

The table shows variation in blade parameters along the radius. It can be observed that blade angle decreases and velocity increases towards the tip.

C. Airfoil Sections

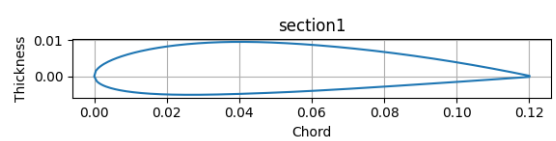

Fig. 2. Airfoil Section at Hub

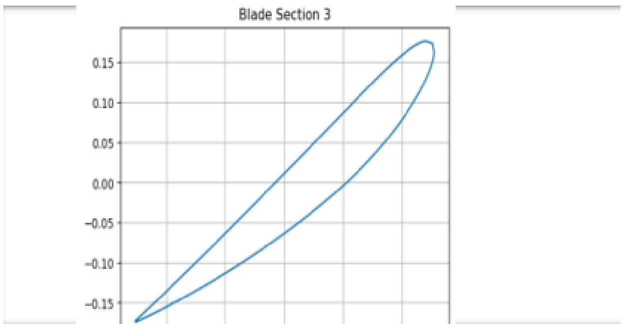

Fig. 3. Airfoil Section at Mid-Span

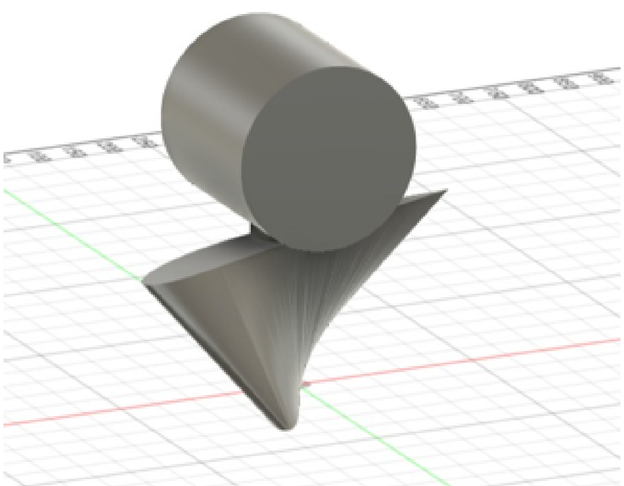

Fig. 4. Airfoil Section at Tip

The generated airfoil sections show gradual variation in geometry. The profiles become thinner and more inclined toward the tip.

D. 3D Blade Model

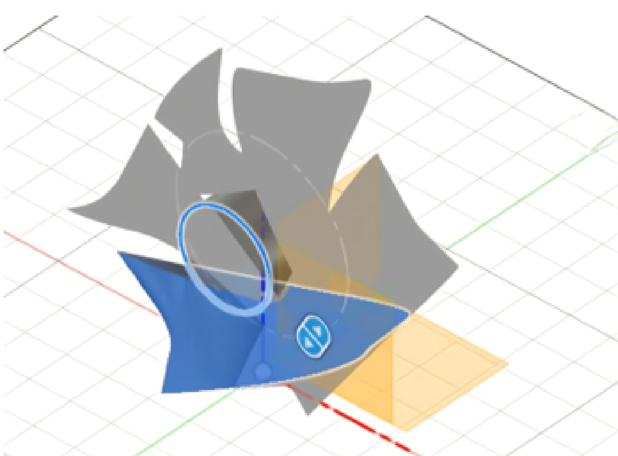

Fig. 5. Kaplan Turbine Blade Model

The 3D blade model shows proper twist and smooth transitions between sections. This confirms the correctness of the computational approach.

V. Design Insights

It was observed that blade twist plays a crucial role in maintaining efficient energy transfer across the blade span. Higher angles near the hub help compensate for lower velocities, while smaller angles near the tip prevent excessive losses.

The reduction in chord length along the radius improves aerodynamic performance and reduces material usage. Additionally, the use of NACA airfoils ensures smooth pressure distribution and minimizes flow separation.

These observations indicate that the computational model successfully captures real-world turbine behavior.

VI. Discussion

The developed approach provides a flexible and efficient method for turbine blade design. By automating calculations and geometry generation, it significantly reduces design time and effort.

The ability to modify input parameters allows engineers to explore multiple design configurations quickly. This makes the model highly suitable for parametric studies and optimization.

However, the current work is limited to geometric modeling. It does not include CFD analysis or structural validation. Incorporating these aspects would further enhance the accuracy and applicability of the model.

VII. Novelty of the Work

The present work introduces a unified computational framework that integrates airfoil generation, hydraulic design, and CAD-ready output within a single workflow. Unlike traditional approaches where these steps are performed independently, the proposed method allows seamless transition from theoretical calculations to practical geometry generation.

The use of Python enables rapid parametric control over turbine inputs such as head, power, and blade count. This allows designers to quickly explore multiple configurations without repeating manual calculations.

Additionally, the direct export of blade sections into CAD-compatible formats eliminates intermediate steps, making the approach highly efficient for real-world applications.

VIII. Engineering Significance

The proposed computational approach has practical implications for turbine design and education. It provides a rapid and accurate method for generating blade geometries without requiring extensive manual calculations.

For students and researchers, the model serves as a valuable learning tool for understanding the relationship between fluid dynamics and blade geometry. For engineers, it offers a preliminary design tool that can be further refined using advanced simulations.

The ability to quickly generate and visualize blade sections makes the approach highly useful for early-stage turbine design and feasibility studies.

IX. Conclusion

A computational framework for Kaplan turbine blade design has been successfully developed using Python. The method integrates airfoil theory, hydraulic calculations, and CAD modeling into a single workflow.

The results demonstrate that the model can generate realistic blade geometries with proper twist and variation. The approach is efficient, adaptable, and suitable for engineering applications.

X. Future Work

Future work may include CFD simulations for performance validation, structural analysis for stress evaluation, and optimization using advanced algorithms. The methodology can also be extended to other types of turbines.

References

[1] J. D. Anderson, Fundamentals of Aerodynamics.
[2] NASA NACA Airfoil Data.
[3] Hydropower Engineering Manuals.